

P54 P3 Flavor and Fermionic CW Gate: Corrected Status

Route-F four-dimensional mainline

5 September 2026

1 Corrigendum and authoritative derivation

This document supersedes the 2 September P3 report. The old conjugation assignment, factor 69.49, proposed $|r| \geq 62.9$ repair target, and empty η_Y profile-domain claim are withdrawn. The full derivations, new stability calculation and constructive repair mechanisms are in `route_f/tex/p54_theory_audit_and_repair.tex` and its PDF. The authoritative numerical ledger is `route_f/output/p54_theory_audit.json`; the compatibility P3 JSON has the same corrected schema.

The declared Yukawas are $h\Psi\Psi\phi^* + f\Psi\Psi\Sigma$. For the implementation's $-iY$ generator, eigenvalue $-i/2$ selects physical $Y = +1/2$. The geometric weights

$$(c_1, c_2, c_3, c_4) = (0.6710153956, 0.7414417126, 0.00005897540343, 0.001588153768)$$

refer to $(\phi_{\text{hol}}, \phi_{\text{anti}}, \Sigma_{\text{hol}}, \Sigma_{\text{anti}})$, not pre-labelled up/down Yukawas. Thus in $Y_d = ah + df$, $Y_e = ah - 3df$, $Y_u = bh + ef$,

$$(|a|, |b|, |d|, |e|) = (c_1, c_2, c_4, c_3), \quad (|r|, |s|) = (1.104954845, 0.033607317),$$

where $r = b/a$, $s = ae/(db)$. The exact identity remains

$$Y_u = \frac{r}{4}[(3+s)Y_d + (1-s)Y_e].$$

Its sharp phase-uniform norm bound gives $y_t \leq 0.00801414124$ at the diagnostic (y_b, y_τ) , versus $y_t = 0.464725012$. This is a frozen-tree-geometry diagnostic, not an unconditional exclusion of the loop-corrected model. SM-only target transport is not the missing Pati–Salam Yukawa matching calculation. No global fit or η_Y likelihood profile has been performed.

2 What the existing loop calculation does not determine

P2 computed $P\Pi_B P$ on the tree light four-plane. It did not compute $Q\Pi_B P$, $Q\Pi_B Q$, or tadpole shifts. For the full copy matrix

$$\mathcal{M}_D^2 = \begin{pmatrix} A & b^\dagger \\ b & C \end{pmatrix}, \quad C > 0,$$

the exact zero condition and vector are

$$A = b^\dagger C^{-1} b, \quad c_{\text{light}} = \frac{c_0 - QC^{-1}b}{\sqrt{1 + \|C^{-1}b\|^2}}.$$

Projected mass retuning alone is not a complete eigenpair condition. The omitted vector rotation is first order in loops, even though the Schur mass correction is second order.

A newly computed tree stability gate gives

$$\lambda_{\text{EFT}} = \frac{1}{6}V^{(4)}[q^4] - \frac{1}{2}J^T C^{-1}J = 0.429206300 - 0.0200527602 = 0.409153539 > 0,$$

where $J_i = V^{(3)}[e_i, q, q]$ and all 290 massive scalar modes relax. Two light directions, two derivative steps, zero-source orthogonality and direct relaxed paths pass 7/7 checks. This supports local tree stability, not global/loop stability or a pole mass.

3 Correct complex heavy-neutrino projection

For $D(z) = \sum_a z_a Y_a$, symmetric invertible fixed M_R , and $z_a = (q_{Ra} + iq_{Ia})/\sqrt{2}$, the exact hard quadratic term is

$$V_F^{(2)} = z^\dagger \Pi z, \quad \Pi_{ab} = -\frac{1}{8\pi^2} \text{Tr}(Y_a^\dagger Y_b W), \quad W = M_R^\dagger M_R \left[\log \frac{M_R^\dagger M_R}{\mu^2} - 1 \right].$$

Its real Hessian retains the imaginary off-diagonal blocks:

$$\Pi_{\mathbb{R}} = \begin{pmatrix} \Re \Pi & -\Im \Pi \\ \Im \Pi & \Re \Pi \end{pmatrix}.$$

For the published Mummidi–Patel H', F' transported with positive-real corrected overlaps, not fitted to this model, the conditional values are

$$\eta_Y = 1.28882418 \cdot 10^{-8}, \quad \|Q(\Pi_F - \delta \xi_F P_{10})Q\|/\omega^2 = 9.68094794 \cdot 10^{-4}.$$

The ratio to the tree gap $0.0305570303 \omega^2$ is 0.03168157. This is not a total boson–fermion certificate or a predicted η_Y . Independent complex finite differences agree at relative error $3.7 \cdot 10^{-8}$. The audit passes 14/14 and overlap extraction 5/5.

4 Next decision

First calculate the complete corrected light eigenpair with a declared tadpole scheme, then couple it to PS Yukawa running and an inverse scalar–flavor feasibility problem. A single scalar η_Y nuisance cannot substitute for the full matrix constraints. If an extension is required, the detailed note gives a PQ-preserving $S^*(16\,16)_{10}\phi/\Lambda$ operator and rank-restricted vectorlike completion. Its explicit third-family toy is not a global fit. A purely antisymmetric 120 repair at fixed geometry instead obeys $y_t - y_c \leq 2U$ and fails the diagnostic target. Minimal Einstein–Cartan torsion does not supply an automatic escape. Neutrino intervals, proton predictions, determinant/portal promotion and a physical Higgs pole remain unclaimed.